C-C-305E
June 16, 1977
SUPERSEDING
Fed. Spec. C-C-305D
October 20, 1971

FEDERAL SPECIFICATION

CHEESE, PASTEURIZED PROCESS SWISS AND PASTEURIZED PROCESS SWISS AND CHEDDAR, OR AMERICAN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for pasteurized process cheese.

1.2 Classification.

1.2.1 Types. Pasteurized process cheese covered by this specification shall be of the following types as specified (see 6.1). Optional labeling for type II and type III shall be as permitted by the Definitions and Standards of Identity for Cheeses and Cheese Products under the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Type I - Pasteurized process Swiss cheese.

Type II - Pasteurized process Swiss and Cheddar cheese.

Type III - Pasteurized process Swiss and American cheese.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

FSC 8910

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

PPP-B-1163 - Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated.
PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standard:

FED-STD-123 - Marking for Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1162 - Sanitary Standards for Cheese and Cheese Products Plants.

(Copies of Military specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

LAWS AND REGULATIONS

US Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US Department of Agriculture

US Standards for Grades of Swiss Cheese

US Standards for Grades of Bulk American Cheese for Manufacturing

US Standards for Grades of Cheddar Cheese

US Standards for Grades of Colby Cheese

US Standards for Condition of Food Containers

(Application for copies should be addressed to the Technical Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

Dairy Plants Surveyed and Approved for USDA Grading Service

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

(Application for copies should be addressed to the Dairy Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

Directory of Meat and Poultry Inspection Program Establishments and Officials

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

Brucellosis Eradication - Recommended Uniform Methods and Rules APHIS-91-1

Uniform Methods and Rules - Bovine Tuberculosis Eradication

(Application for copies should be addressed to the Animal and Plant Health Inspection Service, Veterinary Service, Federal Center Building, Hyattsville, MD 20782.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Public Health Association, Inc.

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street, N.W., Washington, DC 20036.)

National Association of Dairy Equipment Manufacturers

Recommended Practices for Producing Culinary Steam for Processing Milk and Milk Products

(Application for copies should be addressed to the National Association of Dairy Equipment Manufacturers, 1012 Fourteenth St., N.W., Washington, DC 200005.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.1.1 Cheese. All natural cheese used in the preparation of the process cheese shall have been prepared from milk drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the US Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(a) A Modified Accredited Tuberculosis Area in accordance with Uniform Methods and Rules--Bovine Tuberculosis Eradication;
(b) Either

(1) A certified Brucellosis-Free Area; or
(2) Modified Certified Brucellosis Area; or
(3) An area in the process of being certified by the USDA in accordance with USDA Animal and Plant Health Inspection Service Publication 91-1.

When ready for grinding, all natural cheese used to produce the finished product(s) shall be cleaned free from impurities, mold, rind rot, thick dry rind, paraffin, soft spots, salt spots, cheese pests, surface taint or any condition which is unwholesome or which will affect the quality of the finished product. Cheese made from unheated or unpasteurized milk may be used. All cheese shall be subject to inspection.

3.1.1.1 Grades of Swiss, Colby and Cheddar cheese. The interpretation of the requirements (descriptive terms) for grades of natural Swiss, Colby and Cheddar cheese shall be as defined in the US Standards for Grades of Swiss Cheese, US Standards for Grades of Colby Cheese, and US Standards for Grades of Cheddar Cheese, respectively.

3.1.1.2 Swiss cheese, types I, II and III. Natural Swiss cheese used in types I, II, and III shall conform to the requirements of US Grade B or bet .. Pieces, broken slices and ends of cheese may be used for processing provided they are cut from cheeses meeting the above requirements.

3.1.1.3 Cheddar cheese, types II and III. The natural Cheddar cheese shall conform to the requirements of US Grade B or better. Cheddar cheese cured in bulk form, commonly termed barrel cheese, may be used, provided it meets the requirements of US Standard Grade, or better. Pieces, broken slices and ends of cheese may be used for processing provided they are cut from cheeses meeting the above requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:05Z
Check the source to verify that this is the currentversion before use.

3.1.1.4 American Cheese, type III. One or more of the following may be used prepare American cheese to be blended with Swiss cheese (see 3.1.1.2) for pe III.

a. washed or soaked curd
b. granular and stirred curd
c. Colby (see 3.1.1.5)
d. Cheddar (see 3.1.1.3)

e cheese shall conform to the requirements of US Standards for Grades of Bulk erican Cheese for Manufacturing. Pieces and broken slices and ends of cheese y be used for processing provided they are cut from cheese meeting the above quirements. Allowance shall be made for the open body and texture characteris- cs for washed curd, Colby, and granular cheese.

3.1.1.5 Colby cheese, type III. Natural Colby cheese used in type III shall nform to the requirements of US Grade A or better.

3.1.2 Cream (including plastic cream). Cream, if used, shall be sweet, and ee from objectionable odor or flavor. Fresh or frozen cream may be used.

3.1.3 Salt. Salt shall be sodium chloride of food grade quality, free from traneous material.

3.1.4 Emulsifying agents. Emulsifying agents shall be as permitted the Definitions and Standards of Identity for Cheeses and Cheese oducts under the Federal Food, Drug, and Cosmetic Act and regulations omulgated thereunder.

3.1.5 Acidifying and mold inhibiting ingredients. Acidifying and mold hibiting ingredients (when used) shall be as permitted by the Definitions d Standards of Identity for Cheeses and Cheese Products under the Federal od, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.1.6 Water. Water shall meet drinking water standards approved by Federal State authorities. The water may be subjected to suitable chemical or heat eatment. The water shall not impart undesirable odors or flavors to the nished product.

3.1.7 Culinary steam. Steam in direct contact with the products shall be linary steam. Culinary steam shall be free from extraneous material and all be of the quality described in Recommended Practices for Producing linary Steam for Processing Milk and Milk Products. The flavor of the eam condensate shall be bland and free from objectionable flavors and odors. e water for the generation of steam may be treated with boiler compounds which

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

conform to the Food Additives, Amendment of Federal Food, Drug, and Cosmetic Act. Alternatively, a secondary steam generator may be used in which soft water is converted to steam and no boiler compounds are used.

3.2 Preparation. When two or more cheeses of the types described below are used, the natural cheeses shall be blended properly to produce the desired body and texture in the finished product.

3.2.1 Type I. Pasteurized process Swiss cheese shall be prepared by comminuting and mixing one or more lots of natural Swiss cheese as specified in 3.1 into a homogeneous mass with the aid of heat, with or without the addition of salt, water, and with the incorporation of not more than three percent (based upon the weight of the anhydrous form) of a suitable emulsifying agent. Cream may be added to bring the fat content up to the required percentage provided that the fat derived therefrom is less than five percent of the weight of the product. Acidifying and mold inhibiting ingredients may be added.

3.2.2 Type II. Pasteurized process Swiss and Cheddar cheese shall be prepared by comminuting and mixing not more than 75 percent and not less than 51 percent of Swiss cheese and not less than 25 percent of Cheddar cheese into a homogeneous mass as outlined in 3.2.1.

3.2.3 Type III. Pasteurized process Swiss and American cheese shall be prepared as specified in 3.2.2, except for the type of cheese used.

3.2.4 Pasteurization. The process cheese shall be pasteurized by heating to not less than 158°F (70°C) and holding at such temperature for not less than 30 seconds.

3.2.5 Packaging temperature. The process cheese shall be packaged (see 5.1) at not less than 150°F (66°C). After filling and closing, the containers shall be handled and stacked in such a manner as to permit good ventilation and air circulation. The cheese shall be cooled to a temperature of 100°F (38°C) or less within 24 hours from the time the cheese is packaged. Cans may be cooled in water; however, they shall be completely dry prior to packaging into shipping cases.

3.3 Finished product.

3.3.1 All types.

3.3.1.1 Flavor. Type I shall possess a predominant Swiss cheese flavor characteristic of process cheese made from good quality Swiss cheese. For types II and III, the Swiss cheese flavor may be less predominant.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:05Z
Check the source to verify that this is the currentversion before use.

3.3.1.2 Body and texture. The body shall be smooth, medium firm and resilient with practically no gassy pinholes or openings except those caused by entrapped air during filling. The texture shall be close and free from lumps or graininess, slice freely, show no more than slight brittleness or roughness and shall not break when cut. The cheese shall not oil-off when melted.

3.3.1.3 Color. The process cheese shall have a uniform, natural or light straw, pale yellow color.

3.3.1.4 Finish and appearance. The process cheese shall be uniform in shape and size with practically straight edges and flat sides, except for canned cheese. Slices shall separate readily and in whole slices when removed from the primary container. The product, including canned cheese, shall have dry surfaces and be free from mold, cracks, spots crystallized emulsifying agents, or other surface defects. The liner, wrapper, or pouch shall closely contact the cheese on all surfaces and shall entirely enclose the cheese. The wrapper, pouch, or can shall be completely sealed.

3.3.1.5 Analytical requirements. The finished product shall comply with the following analytical requirements (see 4.5.1).

	Type I	Types II and III
Moisture (not more than)	44.0 percent	41.0 percent
Fat (in moisture free substance) (not less than)	43.0 percent	47.0 percent
pH (not below)	5.3	5.3
Salt (not more than)	2.5 percent	2.5 percent
Phosphatase activity	1/	1/

1/ Less than 1 microgram per milliliter of cheese extract.

3.4 Storage, shipment, and delivery temperatures. Finished product to be stored for more than one week prior to shipment shall be cooled promptly to 50°F (10°C) or less and held at this temperature during subsequent storage. The chill room temperature may fluctuate (not to exceed 10°F (6°C)) for short periods to allow for entering the chill room to restock, withdraw, etc. The temperature in the approximate center of the shipping container at time of shipment and delivery shall not exceed 50°F (10°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulagted thereunder.

3.6 Plant qualifications. The product shall originate and be processed, handled and packaged in plants which are approved under the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service and which are listed in Dairy Plants Surveyed and Approved for USDA Grading Service, for the finished product described in this specification. Plants not operating under these regulations shall meet the requirements of MIL-STD-1162. Storage and warehousing facilities for finished product for Military use shall meet the requirements of MIL-STD-1162.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified (see 6.1), the supplier is required to furnish an official inspection certificate issued by the US Department of Agriculture certifying that the grade of the product supplied is as required by the contract or order.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.6.

4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the US Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for Military agencies, (see 4.4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.3.2 Destination inspection. Unless otherwise specified (see 6.3), final inspection shall be made after delivery to the point of destination.

4.4 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1, components and material shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Inspection of dairy products.

4.4.1.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in compliance with the requirements of 3.1.1. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the USDA Animal and Plant Health Inspection Service. (Location of this office can be ascertained by consulting the publication entitled Directory of Meat and Poultry Inspection Program Establishments, and Officials.) Nonconformance to the above requirements shall be cause for rejection of the lot.

4.4.1.2 Examination of cheese component(s). Unless cheese component(s) is accompanied and identified by a USDA grade certificate in accordance with the pertinent grade requirements of 3.1.1.2 through 3.1.1.5, examination shall be made to determine compliance with the flavor requirements of the above referenced paragraphs. Each lot of cheese shall be represented by a cheese vat number and date of manufacture. At least one cheese plug drawn from a representative sample of cheese from each vat (lot) to be used in the specified blends shall be examined to determine compliance with the requirements. Noncompliance of any lot of cheese with the aforementioned referenced flavor shall be cause for rejection of the involved component lot or finished process cheese made therefrom, as applicable.

4.4.1.3 Water. Certified semiannual laboratory test reports from the state or municipality in which the plant is located or from the Federal Government, indicating that the water supply is of a safe sanitary quality, shall be available for review by the Government inspector and may be accepted as evidence of compliance with 3.1.6.

4.4.1.4 Examination of boiler compounds (when applicable). Conformance of boiler compounds to the identity requirements of 3.1.7 shall be ascertained by examination of pertinent labels, invoices or other valid documents. Nonconformance to the above referenced requirements shall be cause for rejection of the involved lot of finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.4.1.5 Examination of cream (including plastic cream), salt, emulsifying, acidifying and mold inhibiting ingredients. Examination of these components, as applicable, shall be made to determine compliance with the requirements of 3.1.2 through 3.1.5 as concerns identity. This determination shall be made by examination of pertinent labels, invoices or other valid documents. In addition, each of the ingredients shall be examined organoleptically, as necessary, to determine conformance to the above referenced requirements and 3.5. The sample for examination shall be approximately 4 ounces (113 g) of ingredient taken from each of five containers or all containers if five or less form a lot. Nonconformance to identity or condition requirements shall indicate an unacceptable ingredient(s) and use of such shall be cause for rejection of the involved end item.

4.4.1.6 Testing of components. Testing of components described in this specification shall be as shown in table I. Results shall be applicable to the lot average. Nonconformance to test requirements shall be cause for rejection of the lot.

TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Measure to nearest	Test ref and rqmt paragraph
Cans and lids 1/	One can and lid	Cans	S-1	Tinplate	0.01 lb (5 g/m^2) per base box	4.5.4 5.1.1.2

1/ Exterior can coating and can label shall be tested in accordance with the applicable subsidiary specification except that tests shall be performed on the same cans submitted for tinplate testing.

4.4.2 In-process examination. Unless otherwise specified (see 6.1), examination shall be performed during processing in accordance with 4.4.2.1 and 4.4.2.2.

4.4.2.1 Preparation and controls. The natural cheese components meeting the applicable flavor requirements shall be examined prior to grinding or blending to determine compliance with the requirements of 3.1.1 and with the preparation requirements of 3.2, 3.2.4 and 3.2.5 as concerns procedure, processing, packaging, handling, and temperature and time intervals. Records of all applicable control data shall be maintained and made available for review.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

ncompliance with one or more of the above referenced requirements, reflected actual examination or by examination of records, shall be cause for rejection the involved quantity of finished product.

4.4.2.2 Formulation. Examination shall be made during formulation of e product(s) to determine compliance with the requirements of the ap- icable product blends and use of permitted components as concerns kind d amounts specified (see 3.2.1 through 3.2.3). Formulation records all be maintained. Noncompliance with the above referenced requirements, determined by examination of records or actual examination shall be use for rejection of the involved quantity of finished product.

4.4.3 Examination of end item. Classification of defects found during amination shall be made in accordance with tables III through V and 4.3.1 through 4.4.3.5. The examination criteria shall be as shown in ble II. The acceptable quality levels (AQLs) for table III shall be pressed as percent defective and as defects per hundred units for tables and V.

TABLE II. End item examination criteria (Military agencies)

ble	Inspection level	Sample unit	Lot size expressed in	AQLS Major A	Major B	Minor
I	S-3	One primary container	Primary containers	-	-	2.5
	I	One primary container	Primary containers	1/	1.5	2.5
	S-2	One primary container	Primary containers	1/	1.5	4.0

/ The finding of any Major A defect shall be cause for rejection of the involved lot of finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight more than one percent under specified weight (for units over 5 pounds (2.27 kg). (Report weight to nearest 1/2 ounce (10 g)).
202	Net weight more than two percent under specified weight (for units over 1 pound (0.45 kg) and up to and including 5 pounds (2.27 kg)). (Report weight to nearest 1/4 ounce (10 g)).
203	Net weight more than five percent under specified weight (for units 1 pound (0.45 kg) or under). (Report weight to nearest 1/8 ounce (g)).

1/ Lot shall be rejected if sample data indicate lot average net weight is less than specified net weight.

TABLE IV. Examination of filled pouch and the over-package

Category			Defect
Major A	Major B	Minor	
			Pouch
101			Tear, hole or not completely sealed.
	151		Pouch material not adhering very tightly to the product surface.
	152		Not clean.
			Over-package
		201	Tear, hole or not completely sealed.
	153		Not clean.
	154		Nomenclature missing, incorrect or illegible.
		202	Labeling information other than nomenclature missing, incorrect or illegible.
	155		Missing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:05Z
Check the source to verify that this is the currentversion before use.

TABLE V. Examination of product characteristics 1/

Category			Defect
Major A	Major B	Minor	
101			Presence of foreign material, e.g., dirt, insect(s), insect part(s), wood, paint, glass, metal, etc.
102			Presence of objectionable flavor or odor, e.g., oxidized, bitter, fruity, yeasty, sour, putrid, disinfectant, etc.
			Flavor
	151		Does not possess a Swiss cheese flavor characteristic of process cheese made from good quality Swiss cheese. 2/
			Body and texture
	152		Not smooth, medium firm or resilient.
	153		Presence of gassy pinholes or openings. 3/
	154		Texture not close and free from lumps or graininess.
		201	Does not slice freely or breaks when cut.
		202	More than slight brittleness or roughness.
	155		Oils off when melted.
			Color
		203	Does not have a uniform, natural or light straw, pale yellow color.
			Finish and appearance
	156		Not uniform in shape and size or shows distortion or lack of conformity to the package shape or has uneven, wet (sticky) surfaces.
	157		Presence of mold, cracks, spots of crystallized emulsifying agents or other surface defects.

A standard cheese trier plug of cheese at 70° to 75°F (21° to 24°C) shall be drawn from each sample unit (see 4.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

2/ For types II and III exception, see 3.3.1.1.

3/ Opening caused by entrapped air during filling shall not be scored as a defect.

4.4.3.1 Examination for packaging, liners and interior coating of cans. Examination for packaging, liners and interior coating of cans shall be made in accordance with the applicable examination criteria of PPP-C-29.

4.4.3.2 Examination of can labeling. Examination of can labeling shall be in accordance with examination criteria of MIL-L-1497.

4.4.3.3 Examination of exterior can coating. Examination of cans for coating when required shall be in accordance with the examination criteria of TT-C-495.

4.4.3.4 Examination of shipping containers. Examination shall be made in accordance with the examination criteria contained in the appendix to PPP-B-636 or PPP-B-1163, as applicable, with the following defects added to the table of examination; Major: Marking missing, incomplete or illegible. Minor: (1) Net weight of carton more than 50 pounds (22.68 kg) as indicated by marked net weight, (2) Arrangement not as specified. When applicable, containers reinforced with other than pressure-sensitive adhesive, filament reinforced tape shall not be acceptable. For level C pack the aforementioned criteria shall be applicable except that only the defects pertaining to marking, arrangement and closure shall apply.

4.4.3.5 Examination of temperature during storage, time of shipment and delivery. Examination shall be made to determine compliance with the requirements of 3.4. Records of storage temperatures shall be maintained during storage and at time of shipment and receipt. The sample unit for this examination shall be one filled and closed shipping container. Lot size shall be expressed in terms of the sample unit and the sample size shall be the number of sample units indicated by inspection level S-2. Nonconformance to any temperature requirements reflected by actual examination or by records shall be cause for rejection of the lot.

4.4.4 Sampling procedure and acceptance criteria for testing of finished product.

4.4.4.1 Sampling cheese for analytical testing. A 1 pound (0.45 kg) composite sample shall be formed by taking a proportionate number of cheese plugs from each of the sample units examined for product characteristics. In all cases the sample container shall be filled to capacity to minimize evaporation or condensation until the cheese is tested. Alternatively, proportionate samples of hot cheese may be taken from the processing line to form the 1 pound (0.45 kg) sample unit for each inspection lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.4.4.2 Testing for analytical requirements. The finished product shall be tested for the characteristics specified in 3.3.1.5. Procedures for testing shall be in accordance with 4.5. The results shall be reported as follows: Moisture, fat (in moisture-free substance) and salt -- to the nearest 0.1 percent, pH -- nearest 0.1 unit and phosphatase on pass or fail basis. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

4.5 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses. Chemical analyses, unless otherwise specified (see 6.1), shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Dairy Products, Section: Cheese.

Test	Method
Moisture	Method I
Fat	Fat
Salt	Total chlorides
pH	Electrometric Method using the Quinhydrone electrode 1/ 2/

1/ Electrometric Method using the Quinhydrone electrode: Grind 2 to 3 grams of cheese with about 0.050 grams of Quinhydrone in a mortar and pestle, and determine the pH with saturated calomel-gold electrodes without any dilution, after adjusting the temperature to 25° C or correcting to that temperature. Quinhydrone is not considered a diluent.

2/ The Electrometric Method using the glass electrode A.O.A.C. method may be used optionally but in case of dispute the method given in footnote 1/ above shall be final. A.O.A.C. Chapter 10: Beverages: Malt Beverages and Brewing Materials, Section: Beer, Method: Hydrogen-Ion Concentration; Electrometric Method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.5.1.1 Phosphatase. Compliance with the phosphatase requirement shall be determined by the Scharer Rapid Phosphatase Test as outlined in Standard Methods for the Examination of Dairy Products, Chapter: Phosphatase Methods to Determine Pasteurization.

4.5.2 Grading. The processed cheese shall be at a temperature of 70°F to 75°F (21° to 24°C) when examined by the Government inspector for flavor, body, texture and color.

4.5.3 Weighing test. Each shipment or lot of processed cheese may be subject to a weighing test by the procuring agency or duly authorized representative.

4.5.4. Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level F or Commercial practice as specified (see 6.1).

5.1.1 Level B.

5.1.1.1 Wrapped and cartoned cheese. The product, in units of 8 ounces (227 g) 1 pound (0.45 kg), 2 pounds (0.91 kg), or 5 pounds (2.27 kg), as specified (see 6.1), shall be completely sealed in the original pouch into which it was filled and the pouch material shall adhere very tightly to the product surface. Commercial packaging which meets this requirement is acceptable. The pouch shall be over-packaged in a commercial carton.

5.1.1.2 Canned cheese. Seven pounds (3.18 kg) of the product shall be packaged in an open-top style, round metal can with soldered side-seam and compound-lined, double seamed ends. The cans (approximate size shall be 603 by 700) shall be made throughout from commerical 0.25 pound (6 g/m^2) electrolytic tinplate per base box and shall be coated inside with an enamel finish furnished by the can manufacturer in accordance with the best commercial practice for the product. The cans shall be fully lined (top, bottom and body) with vegetable parchment or a suitable parchmentized or grease-proof wet-strength paper. The cans shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first consuming activity. The supplier may use his standard practice when it meets this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

5.2 Packing. Not more than 50 pounds (22.6 kg) of the wrapped and carton cheese or six cans of the canned cheese arranged three in length, two in width and one in depth, shall be packed into a snug-fitting shipping container in accordance with level B or C as specified (see 6.1). Style RSC and FOL containers may be top opening or end opening. End opening containers for level B shipment shall have flutes parallel to the score line of the opening.

5.2.1 Level B. The shipping container shall be a fiberboard box, constructed, closed and reinforced in accordance with style RSC or FOL, V3c, V3s or V4s of PPP-B-636 except that two reinforcing bands may be used, one lengthwise and one girthwise. Alternatively, closure of the four flaps on each of the two opposite faces of the style RSC box may be accomplished by making the flaps along the longer edge of the box opening the inner flaps, and those along the shorter edge the outer flaps when in the closed position. Alternatively, the shipping container shall be constructed of wax-impregnated fiberboard in accordance with class I, type SWCFI, grade 275 of PPP-B-1163. The wax-impregnated container shall be closed with hot melt adhesive in accordance with the appendix of PPP-B-636. The wax-impregnated container shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

5.2.1.1 When specified (see 6.1 and 6.4), the shipping container specified in 5.2.1 shall be reinforced with nonmetallic strapping or pressure sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that for V3c, V3s, or V4s boxes, two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.1.2 When specified (see 6.1 and 6.5), the shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC or FOL, W5c or W5s of PPP-B-636 except that two reinforcing bands may be used one lengthwise and one girthwise. Alternatively, closure of the four flaps on each of the two opposite faces of the style RSC box may be accomplished by making the flaps along the longer edge of the box opening the inner flaps, and those along the shorter edge the outer flaps when in the closed position. Alternatively, the shipping container shall be constructed of wax-impregnated fiberboard in accordance with class I, type SWCFI, grade 200, of PPP-B-1163. The wax-impregnated container shall be closed with hot melt adhesive in accordance with the appendix of PPP-B-636. The wax-impregnated container shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

5.2.2 Level C. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the fiberboard boxes shall be closed in accordance with the appendix of PPP-B-636.

5.3 Labeling and marking.

5.3.1 Civil agencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

5.3.1.1 Unit containers. Unless otherwise specified (see 6.1), any labeling of unit containers that comply with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, is acceptable.

5.3.1.2 Shipping containers. Unless otherwise specified (see 6.1), shipping containers shall be marked in accordance with FED-STD-123. Further, the container shall be marked with the quantity and size of packages in the container, and have the following precautionary marking stenciled or printed on the top of the shipping containers in letters between 1-1/2 inches (38 mm) and 2 inches (51 mm) high, all letters being of the same height:

PERISHABLE - KEEP REFRIGERATED

In addition, such other marking requirements as specified by individual agencies shall be included.

5.3.2 Military agencies.

5.3.2.1 For cheese sealed in a protective pouch and carton. Commercial labeling will be acceptable.

5.3.2.2 Cans. The cans of the product shall be labeled in accordance with MIL-L-1497.

5.3.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the shipping container for wrapped and cartoned cheese shall have the following precautionary marking stenciled or printed on the top of the shipping container in letters between 1-1/2 inches (38 mm) and 2 inches (51 mm) high, all letters being of the same height:

KEEP REFRIGERATED
(40° to 45°F)
(4° to 7°C)
DO NOT FREEZE

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of cheese required (see 1.2).
(c) When chemical analysis is not required (see 4.5.1).
(d) Size of unit required (see 5.1.1.1).
(e) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

(f) When packing in accordance with 5.2.1.1 or 5.2.1.2 is required.
(g) When a USDA inspection or grade certificate is not required (see 4.1.1).
(h) When in process examination is not required by Civil agencies (see 4.4.2).
(i) When labeling other than specified in 5.3.1.1 is required.
(j) When shipping containers are to be marked other than in accordance with FED-STD-123 (see 5.3.1.2).
(k) When preacceptance inspection is required by Civil agencies (see 4.2.1).

6.2 Destination inspection (Civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition and quantity only.

6.3 Based on the condition known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.4 Packing specified in 5.2.1.1 is intended for transfer at sea operations and specific overseas operations.

6.5 Packing specified in 5.2.1.2 may be used when experience or knowledge of the destination and mode of shipment would indicate that a lower grade of board can be used.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activity:

GSA - FSS

Project No. 8910-0325

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 45 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:05Z
Check the source to verify that this is the current version before use.